

Project 1 Learner Guide

Sanlam Life, Wealth and Retirement Data Modernization

Why this project exists

Sanlam's life and savings business brings together policy, client, adviser, claim, payment and retirement-fund records from systems that were built and changed over many years. The same person can appear more than once, product labels can differ between systems, and a claim or payment can point to a policy that is missing from the extract. If the underlying records are not reconciled, management can make poor decisions about pricing, claims, distribution and retirement-fund liquidity.

Your job is to create a reproducible path from messy source files to a trusted warehouse and decision-ready reporting. The useful result is not merely a cleaned file. It is a documented, auditable set of rules that lets another analyst explain where each number came from.

Business unit in plain language

- Life cover pays a defined benefit when an insured event, such as death or disability, meets the policy terms.
- Retirement annuities (RAs) are long-term retirement savings products. Contributions, fund value, withdrawals and fees have different meanings; a contribution is not the same as current assets under management (AUM).
- Wealth products can include investment-oriented products whose values change over time.
- Advisers and distribution channels connect customers to products. Production alone does not show whether a channel is profitable: commission, persistency and claims experience matter too.
- Claims are benefit payments or liabilities. A paid-claims measure is not the same as incurred claims, which also considers outstanding reserves.
- Premiums can be recorded as written, billed, collected or earned. State which one your ratio uses.

Two-Pot retirement context

The project brief describes the South African Two-Pot retirement reform, effective from 1 September 2024. At a high level, eligible new contributions are allocated between savings and retirement components, while pre-reform amounts may remain in a vested component. The extract contains component balances and withdrawal activity so you can examine how the mix and withdrawal velocity change over time.

Treat this as an analytics exercise, not a legal-rule engine. Minimum withdrawal values, frequency limits, tax treatment and exceptions can change or depend on the fund and member circumstances. Do not invent a compliance conclusion from the sample. State any assumption and verify current rules before proposing a production control.

Core analytical questions

1. Portfolio loss ratio: Which product and adviser-channel combinations have the highest rolling 12-month paid claims relative to premium? Separate frequency (how often claims occur) from severity (average claim size). Use a consistent premium basis and label the ratio clearly.
2. Retirement liquidity: How much AUM sits in savings, retirement and vested components? How are savings withdrawals changing, and what would a simple 2x-velocity stress mean for outflows? Use opening balances as the denominator for withdrawal velocity.

3. Channel productivity: Which channels create net premium after cancellations and reversals? Compare commission as a share of net premium and examine 12-month persistency. High gross sales are not necessarily durable, profitable business.

4. Data quality: Which source and anomaly type drive defects? How many records are corrected, quarantined, resolved or dropped? Show the rate as well as the count, because source volumes differ.

Every executive finding should state what changed, why it matters, and what action is supported by the evidence. A single headline finding and recommendation should be visible for each question.

KPI definitions to start from

Rolling 12-month loss ratio - Baseline definition: Claims paid in the last 12 months / premiums in the last 12 months | Important decision: Confirm paid vs incurred claims and collected vs written/earned premium.

Claim frequency - Baseline definition: Number of claims / policies in force | Important decision: Define the in-force date and treatment of multiple claims per policy.

Claim severity - Baseline definition: Total claim amount / number of claims | Important decision: State paid amount, incurred amount, currency and treatment of reversals.

12-month persistency - Baseline definition: Policies still in force 12 months after inception / policies inception | Important decision: Handle policies that have not yet reached the 12-month observation point.

Commission efficiency - Baseline definition: Commission paid / net premium production | Important decision: Define net premium as written premium less cancellations and reversals.

Withdrawal velocity - Baseline definition: Savings withdrawals in the period / opening savings-pot balance | Important decision: Specify period, eligible withdrawals, and how zero or negative opening balances are handled.

Anomaly rate - Baseline definition: Flagged records / records ingested | Important decision: Report by source table and anomaly code.

Cleaning efficiency - Baseline definition: Automatically corrected records / flagged records | Important decision: Do not count quarantined or manually reviewed rows as automatic corrections.

Quarantine resolution rate - Baseline definition: Resolved quarantined records / all quarantined records | Important decision: Include an observation date or cohort so the rate is interpretable.

Data engineering concepts

Profile before fixing. Record row counts, null rates, distinct values, duplicate candidates, date ranges and numeric distributions for every source. Confirm which anomalies actually exist instead of assuming the risk list is complete.

Separate correction from suspicion. A category such as Blue Star, BlueStar and BLUE STAR may be safely normalised after you establish that they mean the same thing. A very large death claim may be valid and should be flagged for review rather than capped. An unmatched negative payment may be an error; a negative payment with a matching reversal reference may be legitimate.

Resolve people carefully. Normalise the synthetic ID string for a deterministic first pass. Use a conservative name-and-birth-date match only as a candidate for review. Keep a reversible crosswalk showing which source IDs were linked to the survivor. Avoid placing identifiers in a dashboard or clear-text audit log.

Make the run idempotent. Re-running the same inputs should not append duplicate facts or duplicate audit records. Use a load-batch identifier, stable source-row keys, deterministic rules and controlled upserts or clean rebuilds.

Account for every row. For each source, reconcile: rows read = loaded + quarantined + intentionally dropped. A rejected row needs a reason code, source filename, load timestamp and a clear route to resolve or reject it.

Warehouse and reporting concepts

A star schema separates descriptive dimensions from measurable facts. Use surrogate keys in dimensions, retain source business identifiers for lineage, and use an explicit unknown member only for legitimate late-arriving dimensions. Do not use the unknown key to conceal an orphan that should be quarantined.

Suggested grains:

- Policy-premium fact: one source premium or payment transaction per row.
- Claims fact: one claim per row at the chosen paid/claim grain; document whether multiple payment events are collapsed.
- Payments fact: one ledger transaction per row.
- Two-Pot fact: one account/member and component snapshot per snapshot date, with withdrawals at a documented event or period grain.

Keep a shared date dimension and use one-to-many, single-direction relationships unless a justified case requires otherwise. Validate Power BI totals against independent SQL control totals. In the quality report, show before/after counts and a lineage map from raw columns to warehouse columns.

Suggested anomaly catalogue

Start with the BRD's codes and add any defects found during profiling:

- A01 duplicate client identity: repeated normalised synthetic ID; merge only under a defensible rule and preserve a crosswalk.
- A02 malformed date: fails every accepted input format; impute only with a documented rule and an is_imputed flag.
- A03 numeric outlier: outside a justified fence. Flag plausible large claims; consider capping only clear entry errors such as a mistyped premium.
- A04 orphan reference: source policy, client or adviser key has no parent; quarantine with the missing key type.
- A05 illogical chronology: policy end before start, claim before inception, or future-dated payment.
- A06 category variant: unmapped product, channel, gender or province label.
- A07 balance inconsistency: component balances do not reconcile to total, or an implausible negative component.
- A08 duplicate transaction: repeated transaction reference or matching policy/date/amount; distinguish true duplicates from repeated instalments.

For every correction, retain table, source row key, field, original value, corrected value and rule code. The audit log should explain the transformation without exposing real personal data.

A practical three-week sequence

Week 1 - Understand and clean: profile extracts, agree on KPI meanings, write the anomaly catalogue, implement testable transformations, generate audit and quarantine outputs.

Week 2 - Model and reconcile: build dimensions and facts, choose Type 1 or Type 2 treatment for changing attributes, enforce foreign keys, add useful indexes, and prove row counts and financial totals.

Week 3 - Explain and defend: build the semantic model and dashboard pages, verify DAX against SQL, document RLS and refresh assumptions, and rehearse concise evidence-based recommendations.

Questions to defend in your final review

- Why is the chosen premium basis appropriate for this question?
- What happens to a legitimate high-value claim under your cleaning rules?
- How can an adviser or policy be traced without exposing identifiers in the dashboard?
- Which defects are corrected, and which are quarantined? Why?
- How do you prove that the load can be repeated without changing totals?
- What conclusion would change if the team's chosen treatment of outliers changed?

Short glossary

AUM: assets under management at a point in time. Earned premium: the portion of premium assigned to the period of insurance coverage provided. Persistency: the share of a cohort that remains in force after a defined period. Quarantine: a holding area for records that cannot safely enter the analytical warehouse yet. Surrogate key: a warehouse-generated key that is separate from a source-system identifier. Two-Pot: the retirement savings, retirement and vested component structure described in the project brief.

Training-data caution

Every record in this pack is fictional. Synthetic identity-like strings are not valid identity numbers. Do not infer a person's identity, contact them, or use this dataset for real decisions. The guide is based on the supplied simulated requirements and does not replace legal, actuarial, tax or regulatory advice.